GEOG 172

Spatial Randomness and Spatial Process

In moving from statistical description to statistical inference, we need to establish and understanding of the null hypothesis of most spatial statistical tests. We also need to build a framework for making those inferences.

Lecture 3-1

THE RANDOM ORCHARD CHALLENGE

Plant those trees…randomly

ACTIVITY

Overview: You have just inherited a rectangular farm and 30 seedlings. Those trees need to be planted!

Purpose: Examine the concept of randomness in the context of spatial analysis and its relation to process

Planting Rules:

- Plant the fruit trees as randomly as possible
- Every tree must be planted inside the farm boundary
- Trees can be planted anywhere inside the boundary
- Once a tree is planted, it cannot be moved

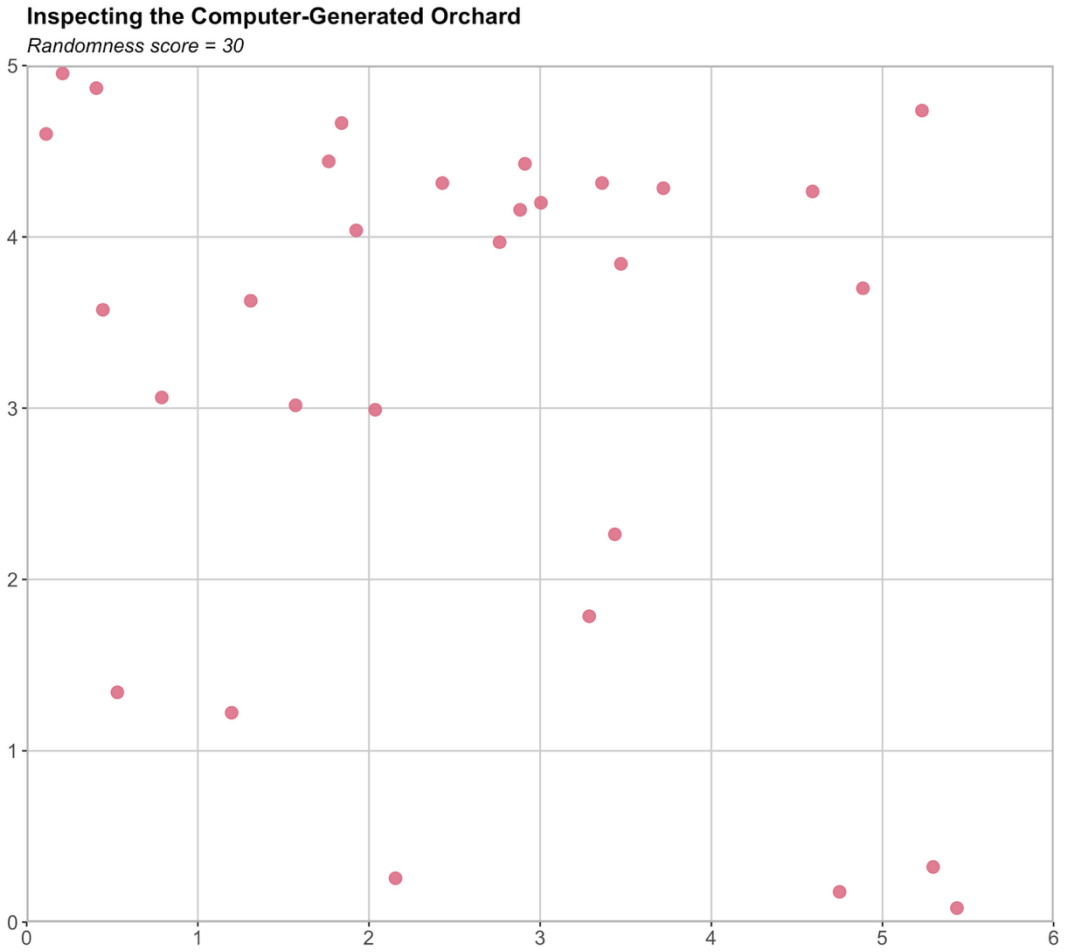
ACTIVITY

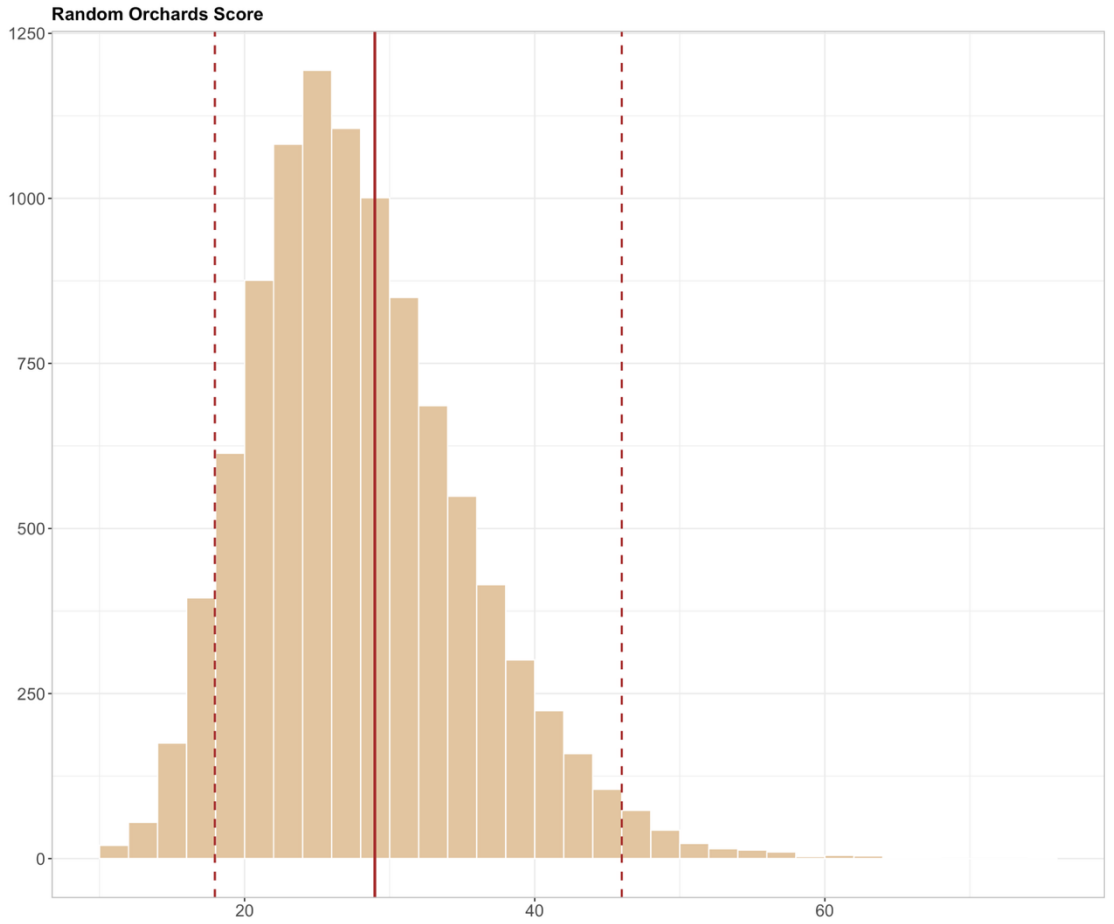
Inspection: Inspection time! You have received a call from the Department of Random Agriculture telling you to perform and report a inspection of neighbor's farm. Use the following rules to evaluate the randomness of your neighboring farm.

Inspection Procedure:

- Connect the tick marks along the edges of the farm to form a grid
- For each grid cell, assign as score to that cell using the table provided to you
- Each point must be assigned to a single cell. If on a grid line, chose a cell
- Sum up the scores for all the cells

ACTIVITY


Inspecting the Computer-Generated Orchard
Randomness score = 30
0
1
2
3
4
5
6


Random Orchards Score
0
250
500
750
1000
1250
20
40
60

COMPLETE SPATIAL RANDOMNESS

(1) Equal Probability - All events have an equal probability of being located in any position.

(1) Independence – The position of any event is independent of the positioning of any other event

FIRST-ORDER SPATIAL VARIATION

Observations across a study region vary from place to place due to changes in the underlying properties of the local environment and the assumption of an equal probability of each area receiving an event cannot be sustained.

- Crime rates may vary because population density varies
- Differences in crop yield may vary because nutrient content in the soil varies

SECOND-ORDER SPATIAL VARIATION

Interaction occur between observations, which creates variation that violates the assumption that event placements are independent of each other.

- Occurrence of a crime may make it more likely that other crimes will occur in an area
- Ash trees share root systems that make trunks grow together

DIRECTIONAL VARIATION

Variation follows a spatial direction. Such a process is called anisotropic and may be contrasted with an isotropic process, where directional effects do not occur.

- Declines in building height as you move away from the center of a city
- Vegetation changes as you move up and down an elevation gradient

STATIONARITY

The rules that govern a process, and control the location of events, or their values, do not change over space.

(1) **First-order Stationarity** – There is no spatial variation in process intensity

(2) **Second-order Stationarity** – There is no interaction between. events

CLUES IN THE NEW ENGLAND WOODS

Reading a forested landscape

ACTIVITY

Overview: You are an ecologist entering a New England woods. Scattered across the forest floor are leaves carrying clues about the processes that have shaped that landscape. You job is to interpret these clues and identify what type of spatial process they might represent.

Purpose: Reinforce the concept of randomness and differentiate between spatial effects. Learn that these concepts can help you explain real landscapes.

Clue Guide:

- In pairs, pick up one **Landscape Clue Leaf**
- Read the clue and decide what type of spatial effect it represents.
- Place your leaf on the forest board, matching the effect you identified



Simulating Spatial Processes

From concept to computer